

Video 4: MECH 5200 Numerical Methods for PDEs : *1-Dimensional Finite Difference Code*

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Quick review of Finite differences

- Taylor series expansions
- Idea: To approximate the derivative of a function using the function value at nearby point(s)
- Idea: The FD approximation will introduce some error vs. the actual derivative
- Today: Let's try this out with a simple function:

$$f(x) = \sin(2\pi x) \quad (1)$$

- What is the first derivative at $x = 0.1$?
- What is the second derivative at $x = 0.1$?
- After: What is the first and second derivative across $x = [0, 1]$?

First Derivative: $\frac{du}{dx} = \frac{u_{i+1} - u_i}{\Delta x} + O(\Delta x)$

- Actual solution $\frac{du}{dx} = 2\pi \cos(2\pi 0.1) = 5.083203692315259$
- **Forward finite difference** approximation:

$$\begin{aligned}\frac{u_{i+1} - u_i}{\Delta x} &\simeq \frac{du}{dx} \\ \frac{\sin(2\pi 0.11) - \sin(2\pi 0.10)}{0.01} &= 4.963873745621649...\end{aligned}$$

- error = $5.08320... - 4.96387... = 0.1193299...$

First Derivative: $\frac{du}{dx} = \frac{u_i - u_{i-1}}{\Delta x} + O(\Delta x)$

- Actual solution $\frac{du}{dx} = 2\pi \cos(2\pi 0.1) = 5.083203692315259$
- **Backward finite difference** approximation:

$$\begin{aligned}\frac{u_i - u_{i-1}}{\Delta x} &\simeq \frac{du}{dx} \\ \frac{\sin(2\pi 0.1) - \sin(2\pi 0.09)}{0.01} &= 5.1958457313476...\end{aligned}$$

- error = $5.08320... - 5.195845... = -0.112642...$

First Derivative: $\frac{du}{dx} = \frac{u_{i+1} - u_{i-1}}{2\Delta x} + O(\Delta x^2)$

- Actual solution $\frac{du}{dx} = 2\pi \cos(2\pi 0.1) = 5.083203692315259$
- **Central finite difference** approximation:

$$\begin{aligned}\frac{u_{i+1} - u_{i-1}}{2\Delta x} &\simeq \frac{du}{dx} \\ \frac{\sin(2\pi 0.11) - \sin(2\pi 0.09)}{0.02} &= 5.079859738484648\ldots\end{aligned}$$

- error = $5.08320\ldots - 5.07986\ldots = 0.00334395\ldots$
- Error is much better than previous estimate!

First Derivative: $\frac{du}{dx} = \frac{3u_i - 4u_{i-1} + u_{i-2}}{2\Delta x} + O(\Delta x^2)$

- Try this out on your own...

Second Derivative: $\frac{d^2u}{dx^2} = \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2} + O(\Delta x^2)$

- Try this out on your own...

Let's use MATLAB to compute finite differences

- Setup MATLAB to compute the first and second derivative at regular intervals.
- How good is the approximation? Error $\rightarrow$ as interval decreases.
- Let's see how a solution might look...

Calculating the Derivatives

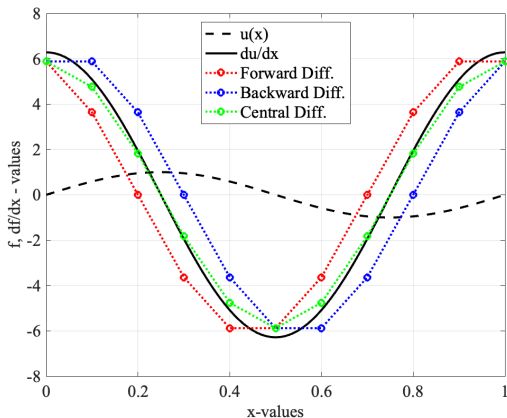


Figure: First Derivative Calculations : $\frac{df}{dx}$ - numerical derivative approximation using equally spaced intervals.

Calculating the Error in the FD Derivative

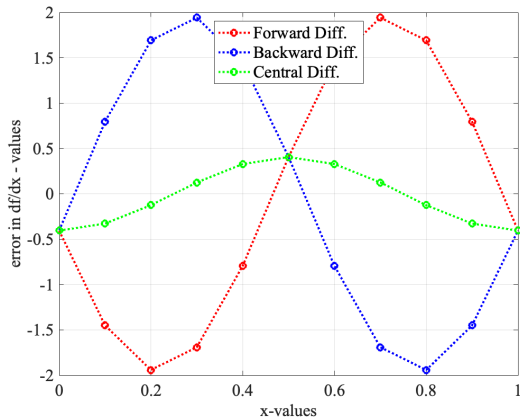


Figure: First Derivative Error Calculations : $\frac{df}{dx}$ - numerical derivative approximation at equally spaced intervals.

Calculating the Derivatives: $\Delta x \rightarrow \frac{\Delta x}{2}$

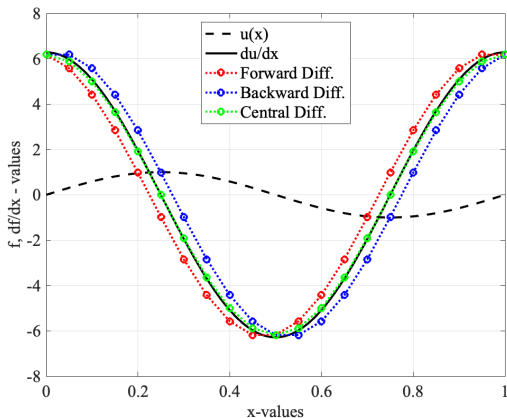


Figure: First Derivative Calculations : $\frac{df}{dx}$ - numerical derivative approximation using equally spaced intervals.

Calculating the Error in the FD Derivative: $\Delta x \rightarrow \frac{\Delta x}{2}$

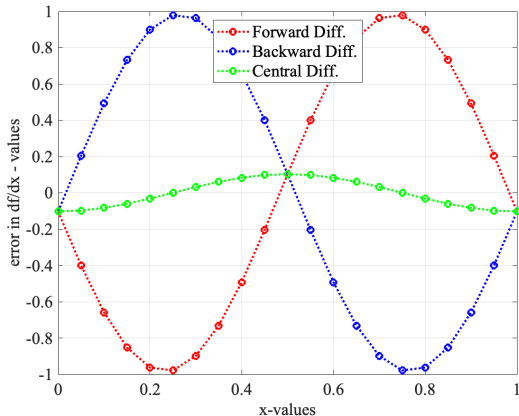


Figure: First Derivative Error Calculations : $\frac{df}{dx}$ - numerical derivative approximation at equally spaced intervals.

Calculating the Second Derivative

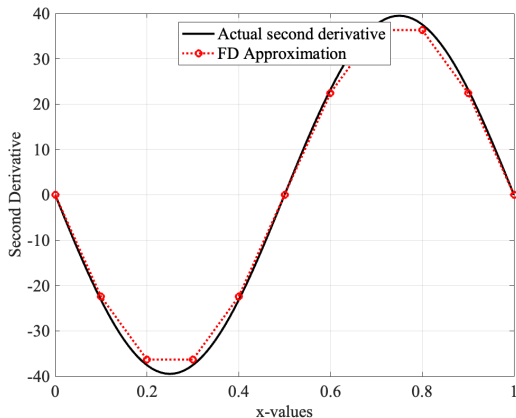


Figure: Second Derivative Calculations: Approximation of $\frac{d^2 \sin(2\pi x)}{dx^2}$ at equally spaced intervals.

Calculating the Error in the Second Derivative Approximation

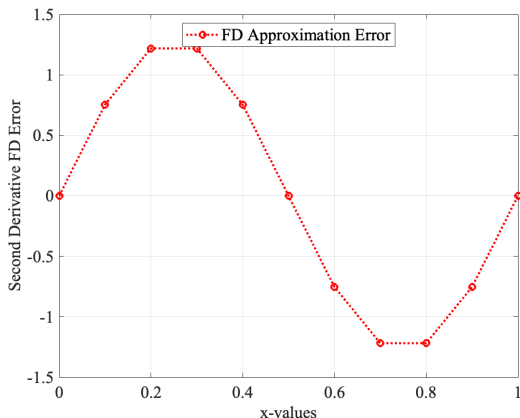


Figure: Second Derivative Calculations: The error in the approximation of $\frac{d^2 \sin(2\pi x)}{dx^2}$ at equally spaced intervals.

Calculating the Second Derivative: $\Delta x \rightarrow \frac{\Delta x}{2}$

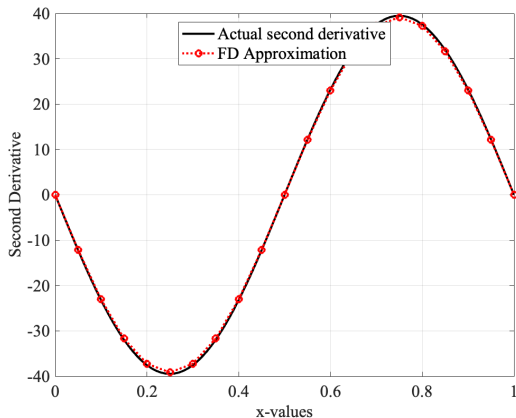


Figure: Second Derivative Calculations: Approximation of $\frac{d^2 \sin(2\pi x)}{dx^2}$ at equally spaced intervals.

Calculating the Error in the Second Derivative

Approximation: $\Delta x \rightarrow \frac{\Delta x}{2}$

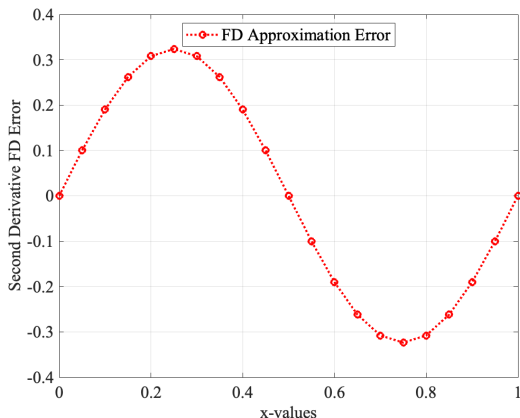


Figure: Second Derivative Calculations: The error in the approximation of $\frac{d^2 \sin(2\pi x)}{dx^2}$ at equally spaced intervals.

How to quantify the error in a meaningful manner?

- Error = Actual Solution - Numerical Solution
- We could use $\max(\text{ActualSolution} - \text{NumericalSolution}) \rightarrow L_\infty$ -norm
- We may want to know the overall distance your approximation is from the actual solution:

$$\|Error\| = \left(\int [u(x) - \hat{u}(x)]^2 dx \right)^{1/2} \approx \sqrt{\sum_{i=1}^N ((u_i - \hat{u}_i)^2 \cdot \Delta x)} \quad (2)$$

- This is considered a modified L_2 norm of the error. Let's see how $\|Error\|_2$ improves as $\Delta x \rightarrow 0$
- Try to determine your error!

Error as a function of Δx

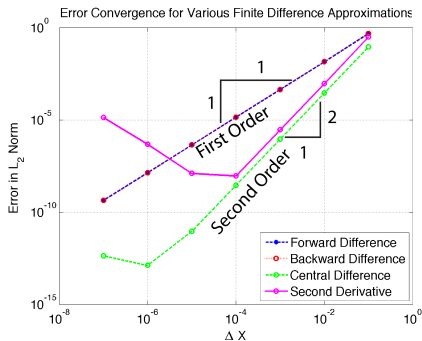


Figure: L_2 -norm of the error (log-log plot)